

Session 11: Linux Networking Lab

Exercise 1 : Discover Your Network Interfaces

Commands

- nmcli
- ifconfig
- ip addr

Questions

1. Which interface is your primary network interface?
2. Does your primary interface have an IPv4 address? Verify using ip addr show
3. Which IPv6 addresses are assigned?
4. What is the status of your device and its active connection ?

Exercise 2 : Disable and Enable a Network Interface

Commands

- ip link set
- ping google.com

Questions

1. Disable your network interface and check using ping (e.g. google.com). What happens?
2. What happens when the interface is up again?

Exercise 3 : Assign a Static IPv4 Address

Your network interface is enp0s3. Assign the following network settings:

- **Static IP:** 172.25.250.11
- **Subnet mask:** 255.255.255.0 (/24)

- **Gateway:** 172.25.250.254
- **DNS server:** 172.25.250.254
- **Connection name:** lab

Your task is to configure your interface with these settings using NetworkManager (nmcli), then verify it.

Exercise 4 : Add a Second IPv4 Address

Assign a second Ipv4 address (10.0.1.1/24) to your “lab” connection and verify that both IPs are on your interface. Why do you think it is useful to have multiple IP.

Exercise 5 : Assign an IPv6 Address

Add IPv6 address (*2001:db8::1/64*) to your interface (e.g. *enp0s3*), for this use *ip*

1. Which command assigns the IPv6 address?
 2. Which command verifies it?
 3. Which IPv6 addresses appear?
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Exercise 6 : Explore Routing

Commands

- `ip route`

Questions

1. Which interface is used to reach the Internet? Try using google.com
 2. How do you remove the temporary route?
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Exercise 7 : Configure /etc/hosts

Assign a custom hostname `private` to the IP address `10.0.1.1`.

Verify that your computer can resolve the hostname and communicate with it using `ping`.

Why?

As seen in class, normally, your computer uses DNS to convert hostnames (like google.com) into IP addresses.

/etc/hosts is a local file that lets your system resolve hostnames without using DNS

Any command using private (like ping private or ssh private) will directly use 10.0.1.1.

This is very fast and works even if DNS is not available

Exercise 8 : Explore /etc/services (5 min)

Questions

1. What ports correspond to SSH and HTTP?
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Exercise 9 : Create a Virtual Interface

Create a virtual network interface , assign an IP, bring it up, and verify its configuration.

Commands

- sudo ip link add link DEVICE name macvtap1 type macvtap mode bridge
 - ip addr
 - sudo ip link set
 - ip addr show
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Exercise 10 : Troubleshoot Connectivity

Questions

1. Is your gateway reachable?
2. Which ports are listening?